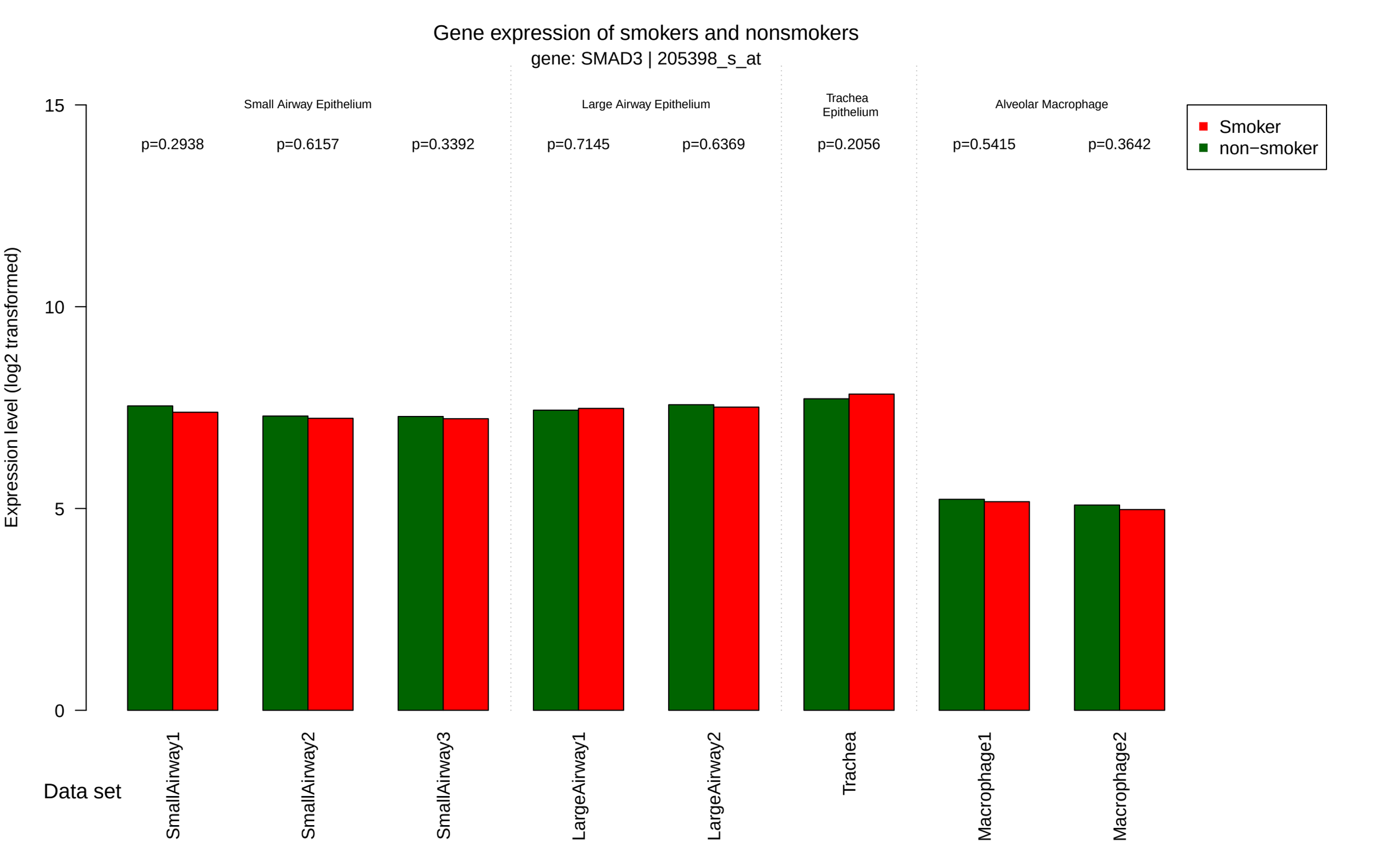
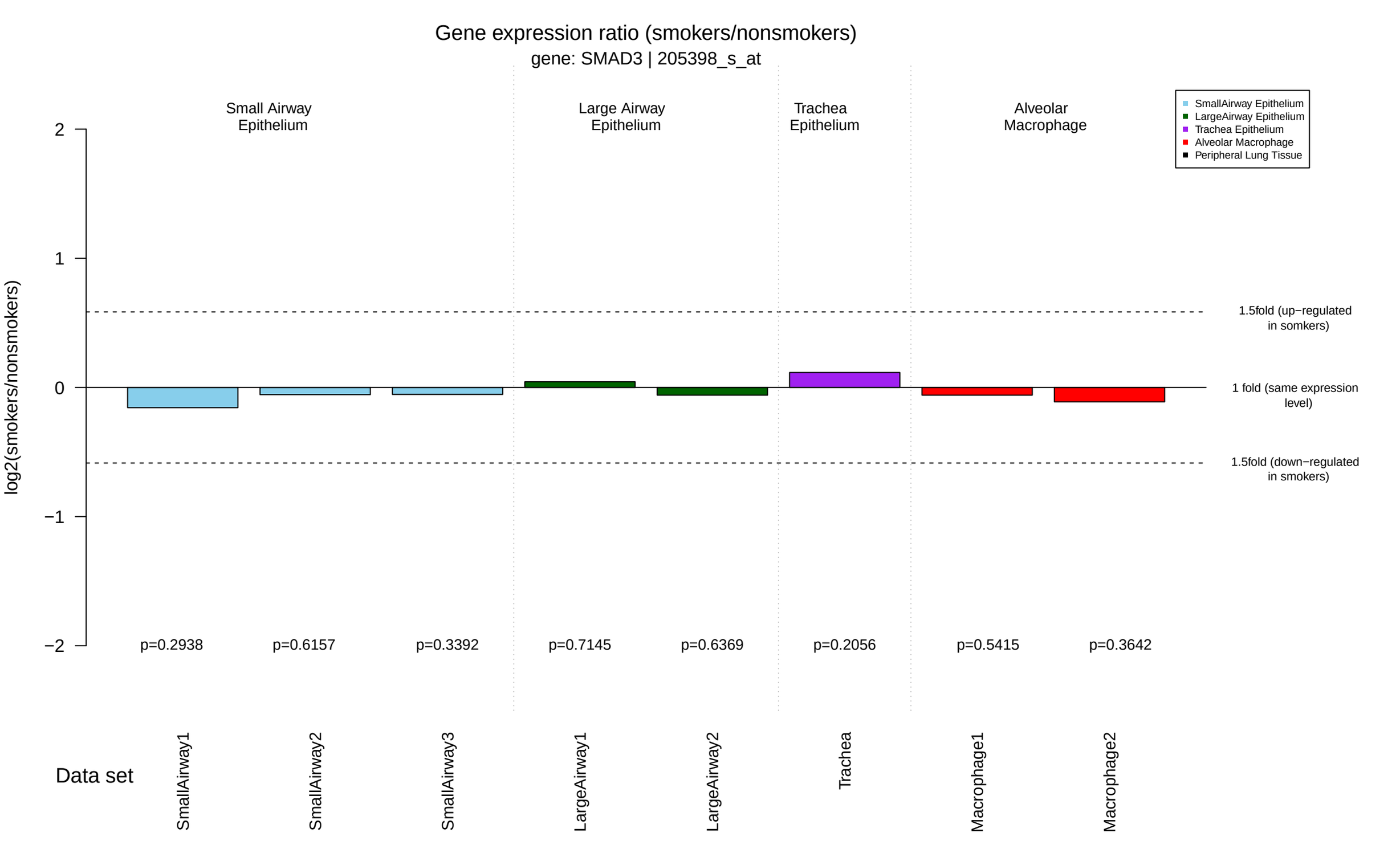
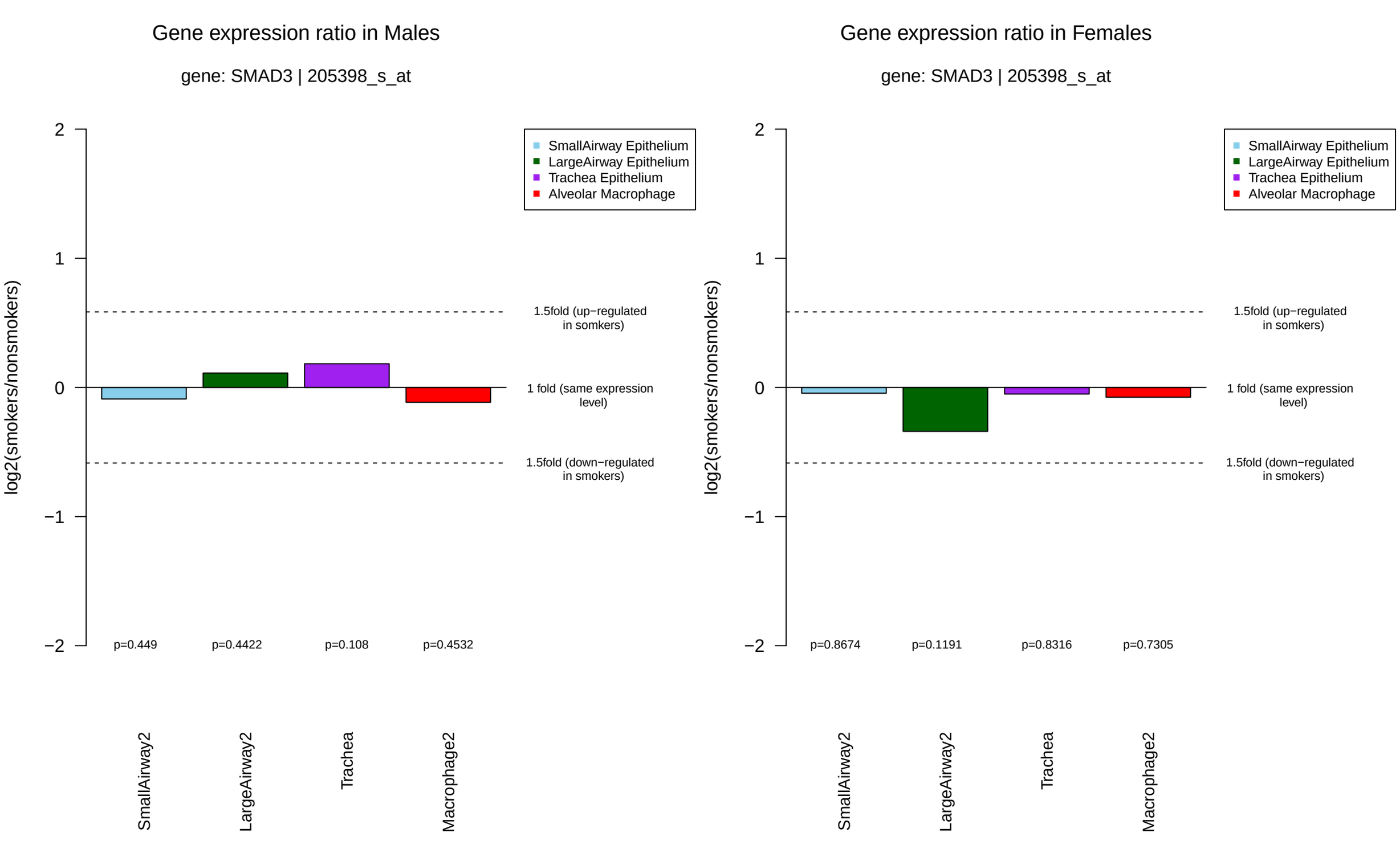
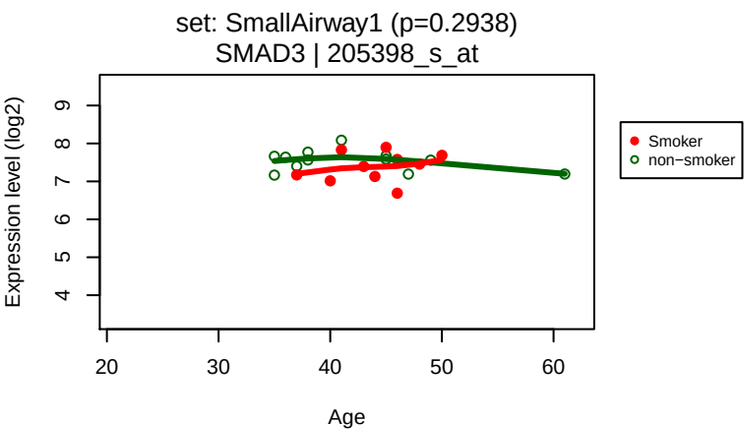
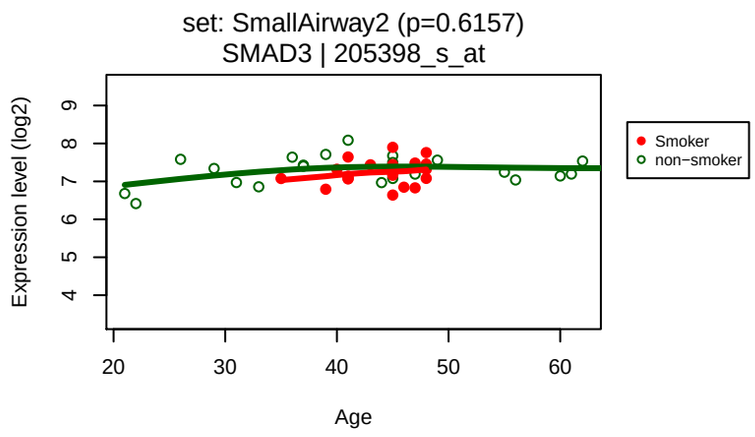
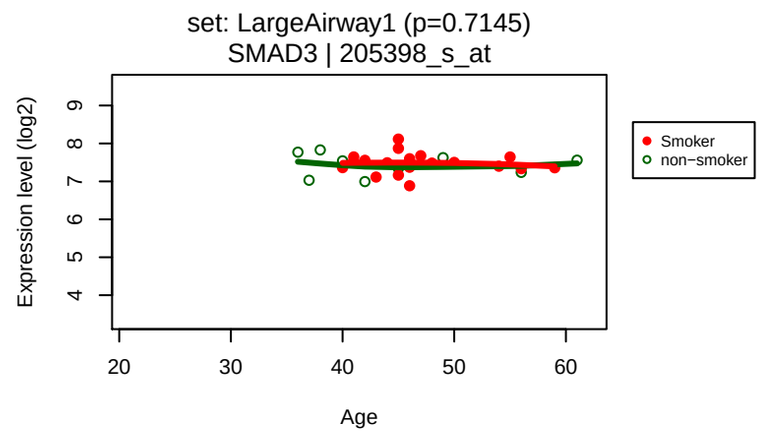

Gene expression of smokers and nonsmokers
gene: SMAD3 | 205398_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2938
p=0.6157
p=0.3392
p=0.7145
p=0.6369
p=0.2056
p=0.5415
p=0.3642
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

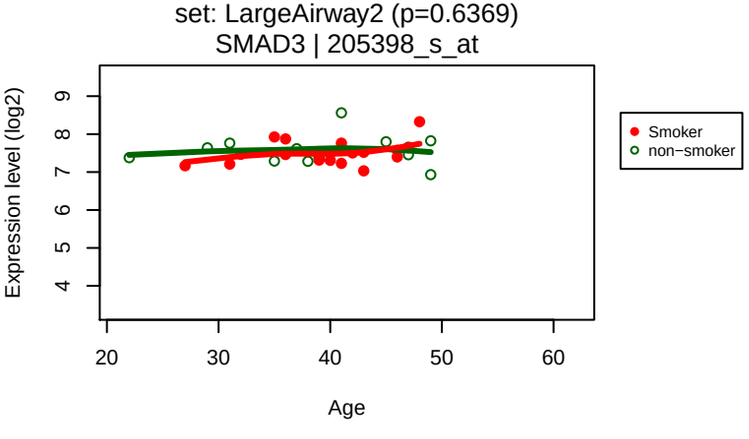

Gene expression ratio (smokers/nonsmokers)
gene: SMAD3 | 205398_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2938
p=0.6157
p=0.3392
p=0.7145
p=0.6369
p=0.2056
p=0.5415
p=0.3642
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

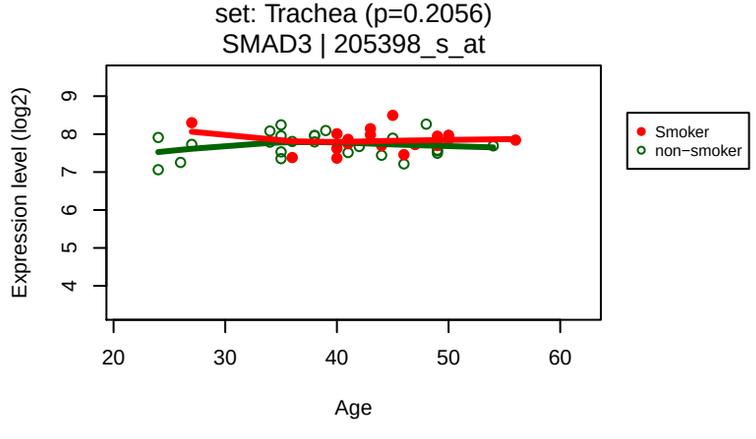

Gene expression ratio in Males
gene: SMAD3 | 205398_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.449
p=0.4422
p=0.108
p=0.4532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD3 | 205398_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8674
p=0.1191
p=0.8316
p=0.7305
SmallAirway2
LargeAirway2
Trachea
Macrophage2

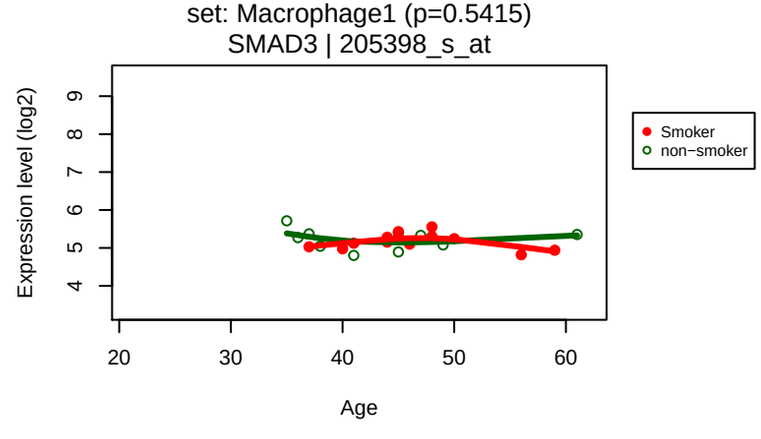

set: SmallAirway1 (p=0.2938)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker

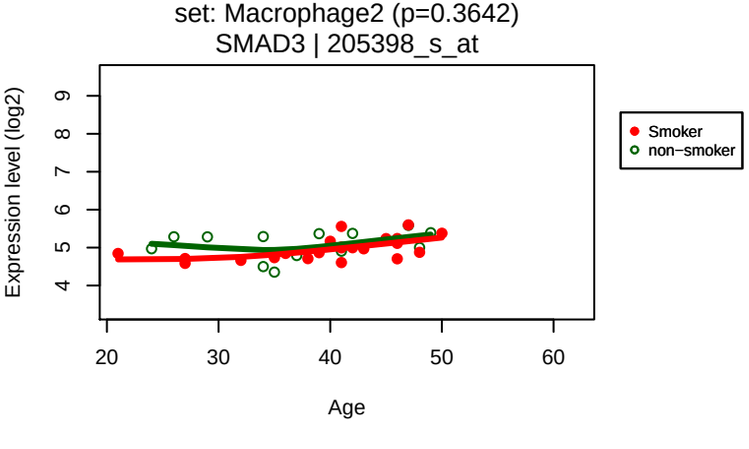

set: SmallAirway2 (p=0.6157)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker

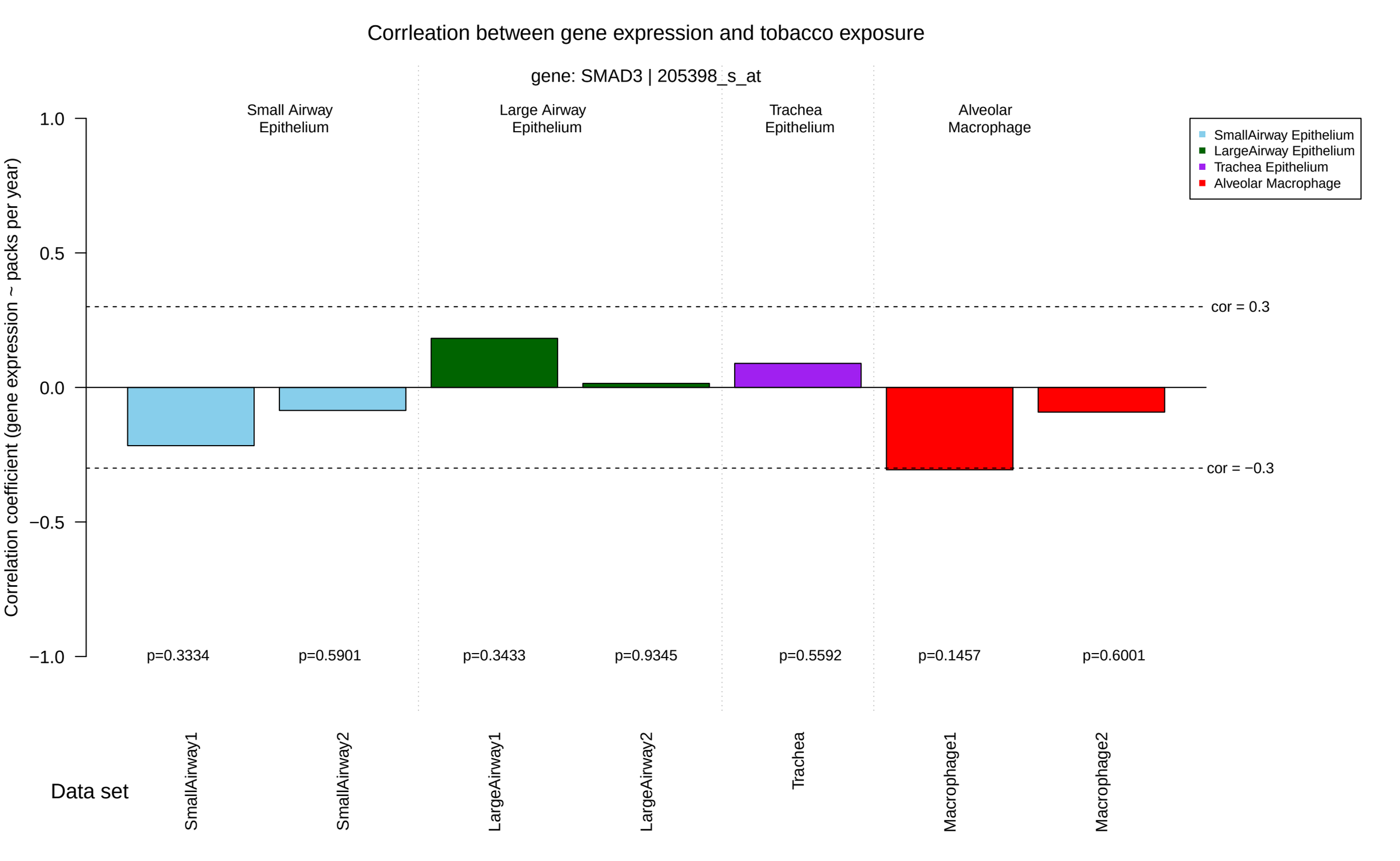

set: LargeAirway1 (p=0.7145)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6369)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2056)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5415)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3642)
SMAD3 | 205398_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD3 | 205398_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3334
p=0.5901
p=0.3433
p=0.9345
p=0.5592
p=0.1457
p=0.6001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2